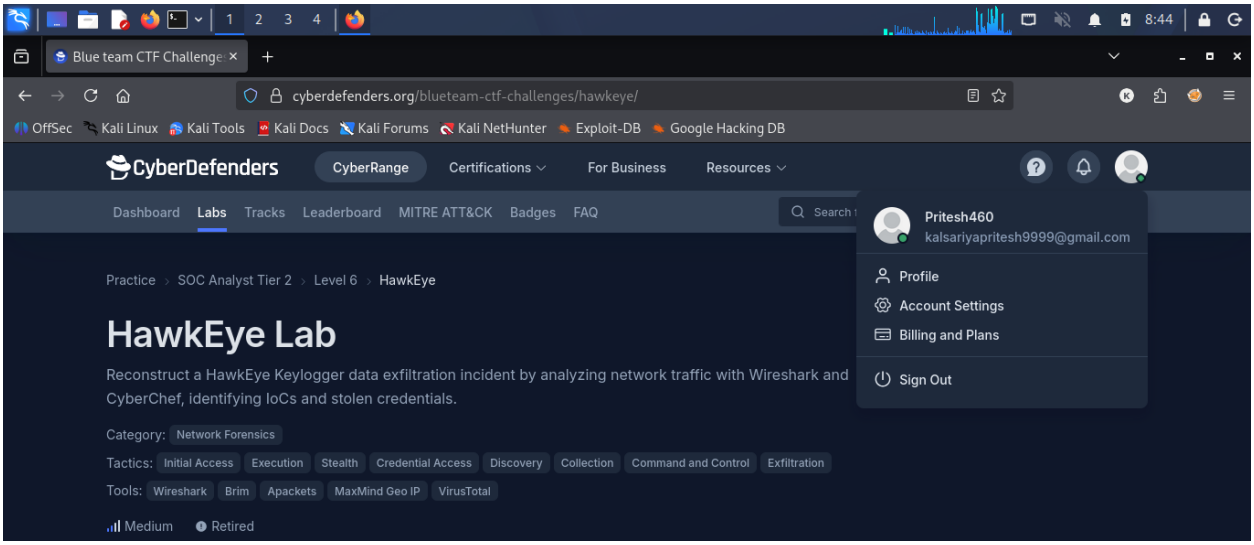
HawkEye Lab: Network Traffic Analysis

1. Objective

The objective of this lab was to investigate the HawkEye challenge from CyberDefenders by analyzing network traffic, identifying suspicious activities, and extracting important information from the captured packets.

2. Lab Setup

1. Opened CyberDefenders – Blue Team CTF Challenges.
2. Created an account and logged in.
3. Opened the HawkEye Lab and downloaded the challenge file.
4. Extracted the downloaded archive.
5. Opened the `stealer.pcap` file in Wireshark for network traffic analysis.



3. Tools Used

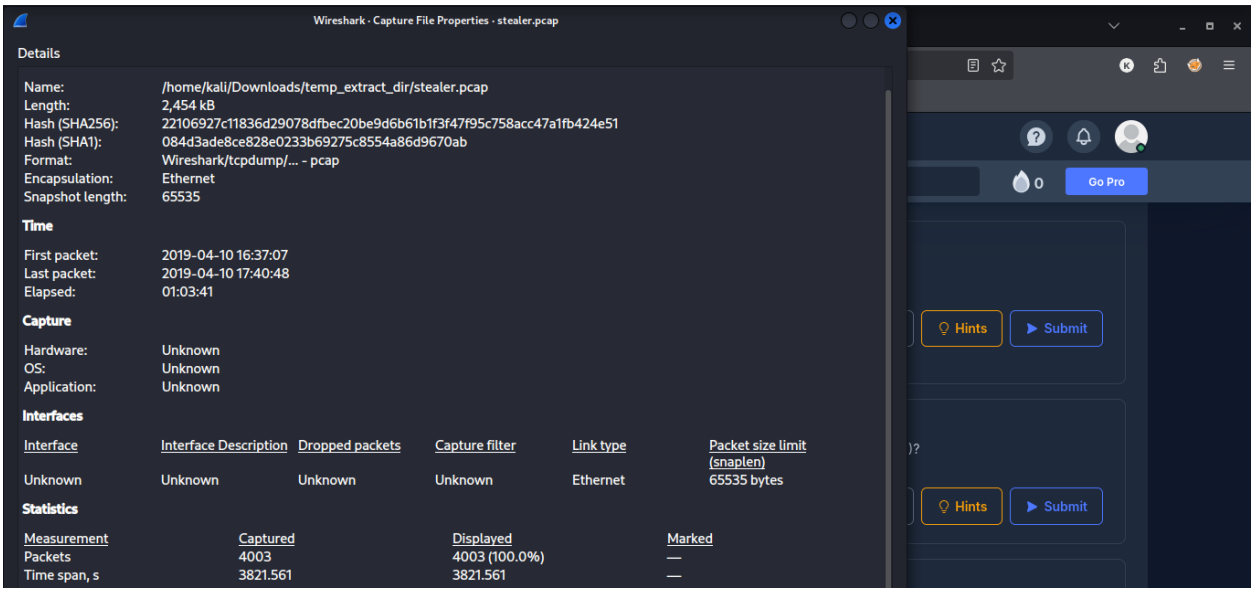
- **Wireshark:** Analyzed network packets, DNS queries, HTTP traffic, and SMTP communication.
- **CyberChef:** Decoded Base64 data and analyzed encoded information.
- **macvendors.com:** Identified the manufacturer of a network interface card (NIC).
- **AbuseIPDB / IP lookup:** Checked IP address information and geolocation.

4. Lab Investigation and Findings

Q1. How many packets does the capture have?

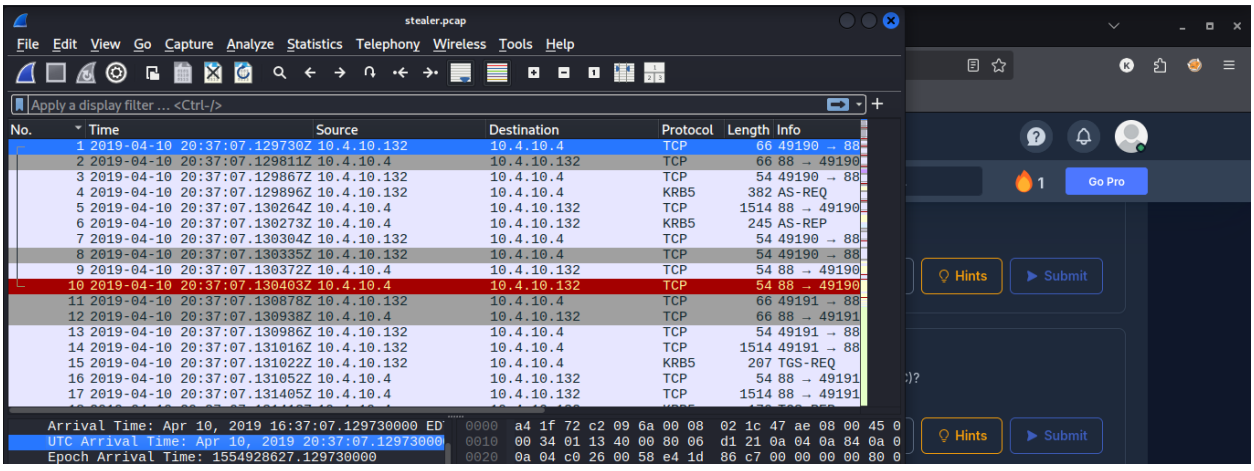
Answer: 4003

Analysis: Statistics → Capture File Properties → Checked the total packet count.



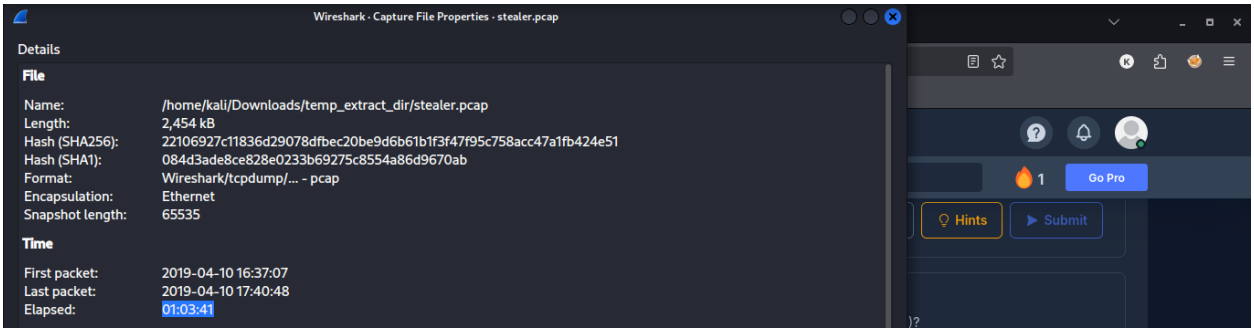
Q2. At what time was the first packet captured (UTC)?

Answer: 2019-04-10 20:37



Q3. What is the duration of the capture?

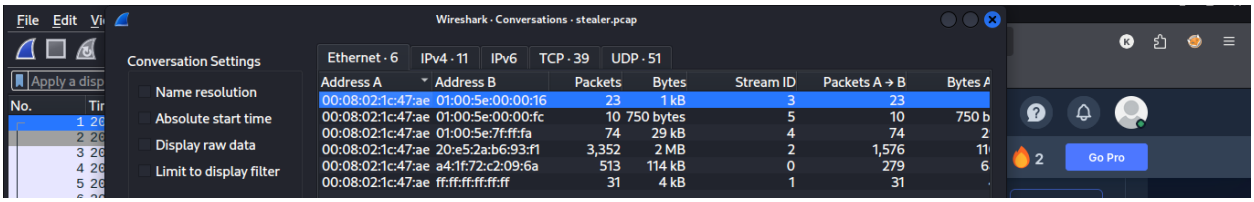
Answer: 01:03:41



Q4. What is the most active computer at the link level?

Answer: 00:08:02:1c:47:ae

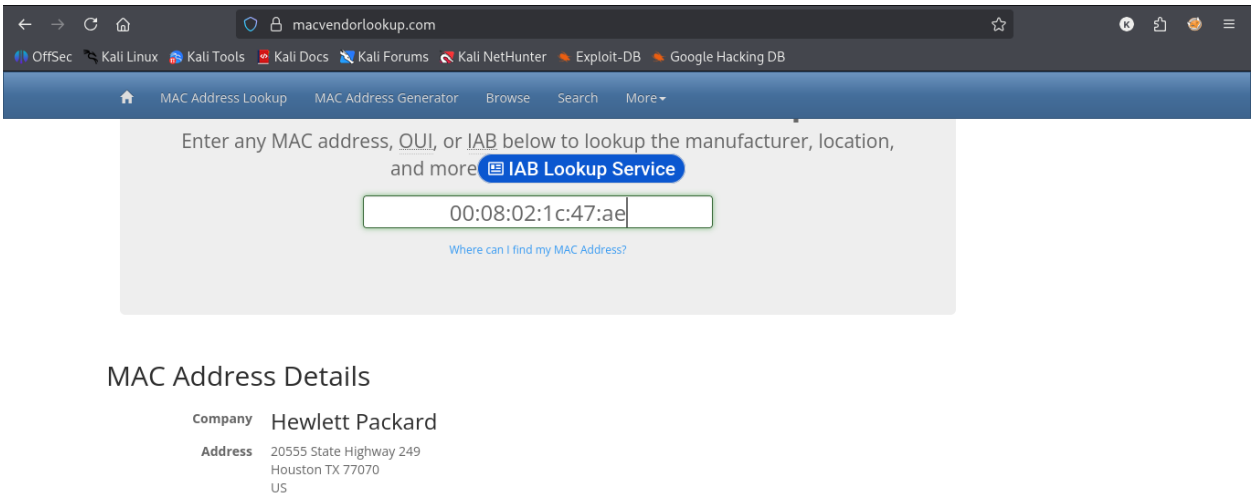
Analysis: Statistics → Conversations → Ethernet → Identified the most active MAC address.



Q5. Manufacturer of the NIC of the most active system at the link level?

Answer: Hewlett-Packard

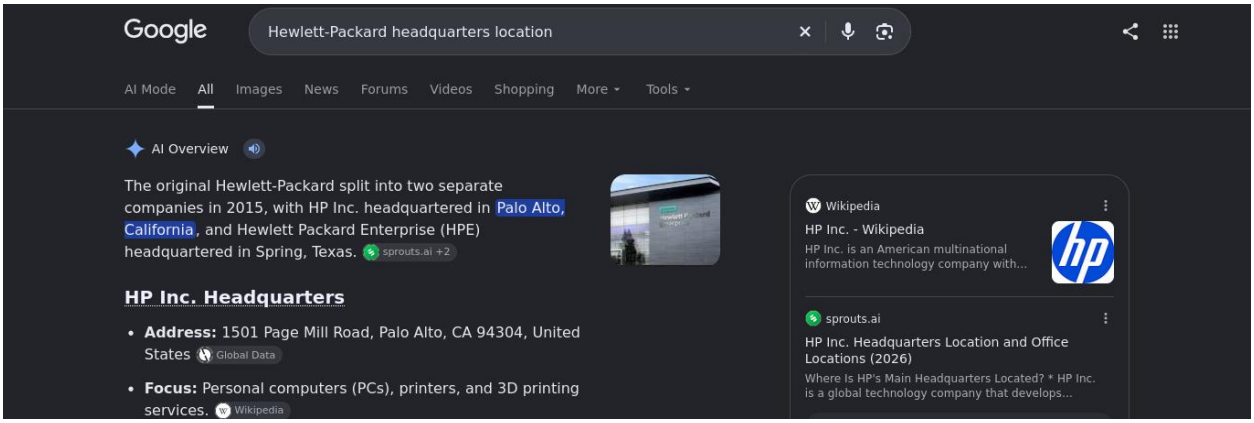
Analysis: Opened macvendorlookup.com → Entered the MAC address → Checked the manufacturer.



Q6. Where is the headquarters of the company that manufactured the NIC?

Answer: Palo Alto

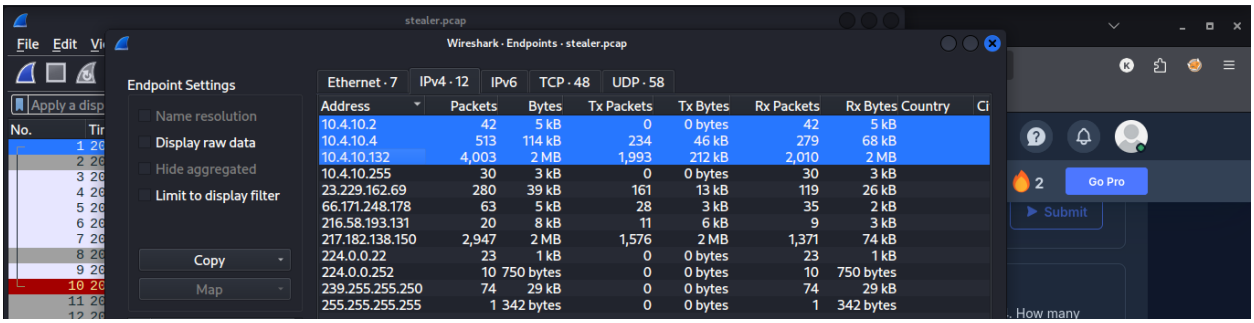
Analysis: Searched for Hewlett-Packard headquarters → Checked the location.



Q7. The organization works with private addressing and netmask /24. How many computers in the organization are involved in the capture?

Answer: 3

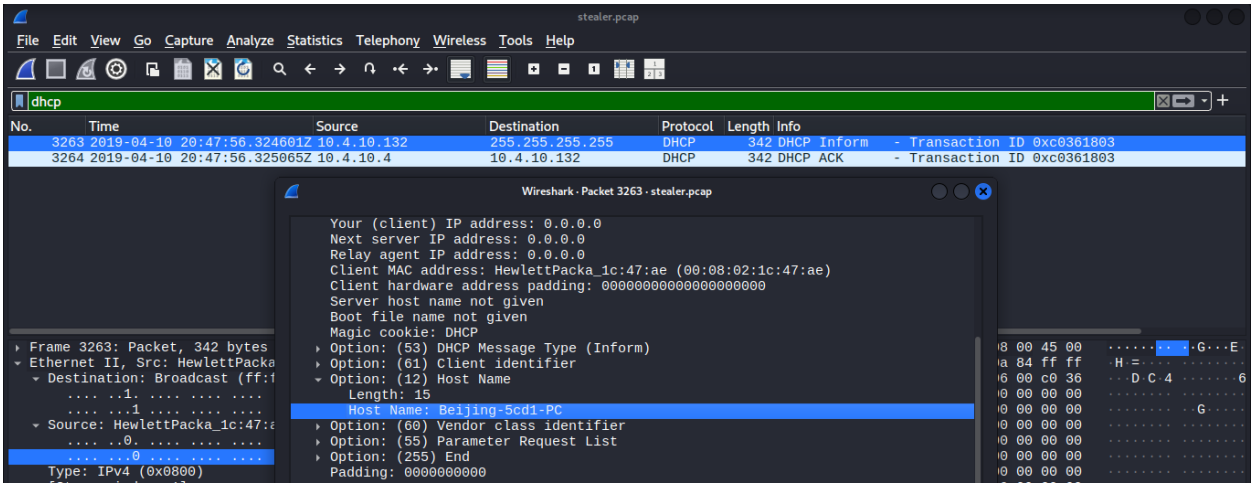
Analysis: Statistics → Endpoints → Checked the private IP addresses.



Q8. What is the name of the most active computer at the network level?

Answer: BEIJING-5CD1-PC

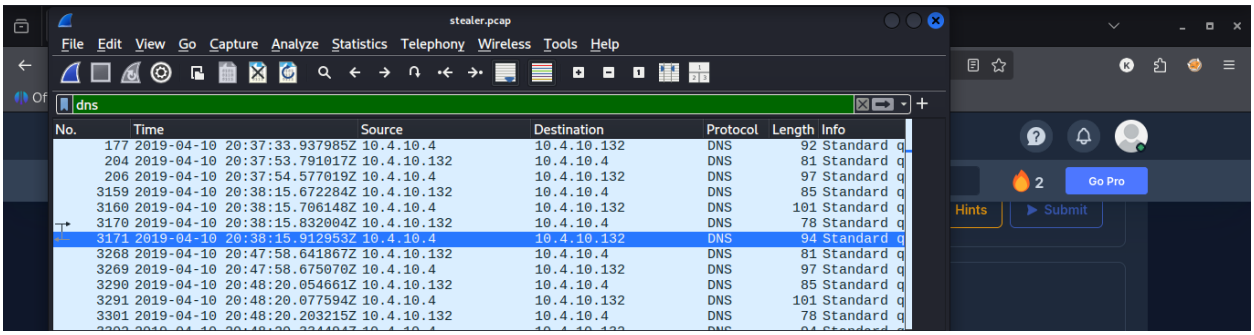
Analysis: Applied the DHCP filter → Checked the DHCP packet details → Identified the hostname.



Q9. What is the IP of the organization's DNS server?

Answer: 10.4.10.4

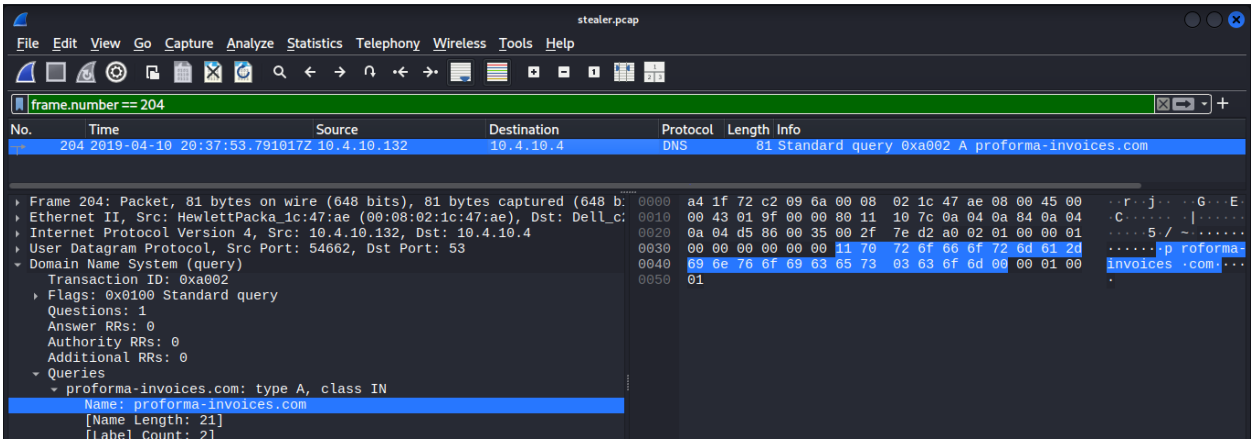
Analysis: Applied DNS filter → Checked DNS packets → Identified the DNS server IP.



Q10. What domain is the victim asking about in packet 204?

Answer: proforma-invoices.com

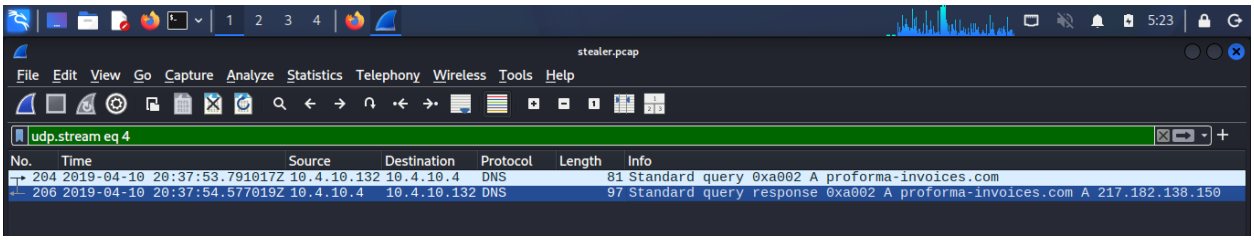
Analysis: Applied `frame.number == 204` filter → Checked DNS query → Found the domain name.



Q11. What is the IP of the domain in the previous question?

Answer: 217.182.138.150

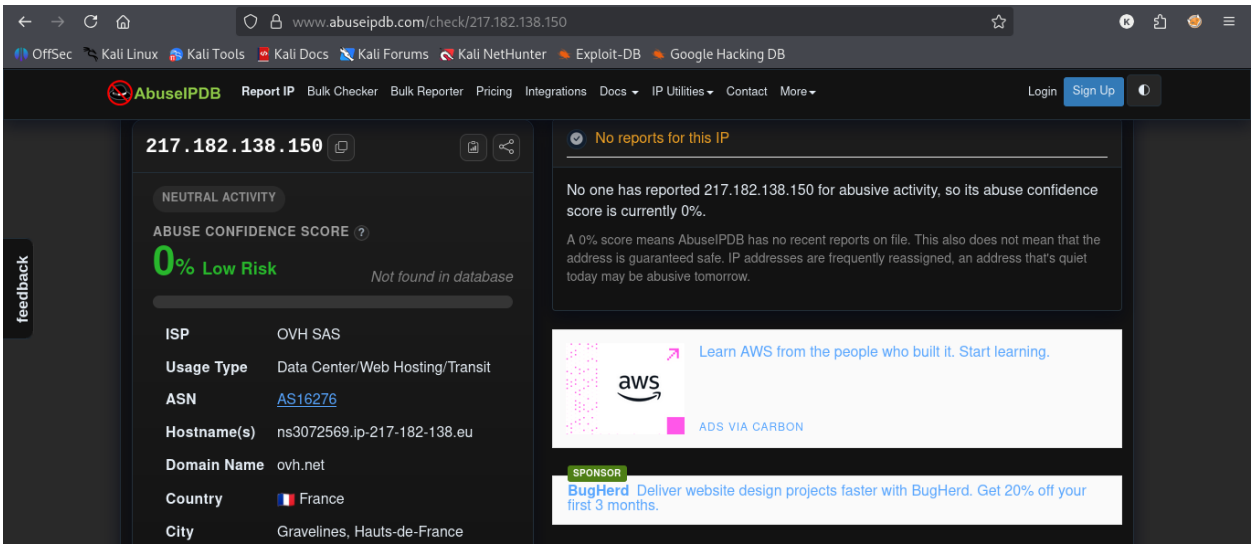
Analysis: Followed UDP Stream → Identified the resolved IP address.



Q12. Indicate the country to which the IP in the previous question belongs.

Answer: France

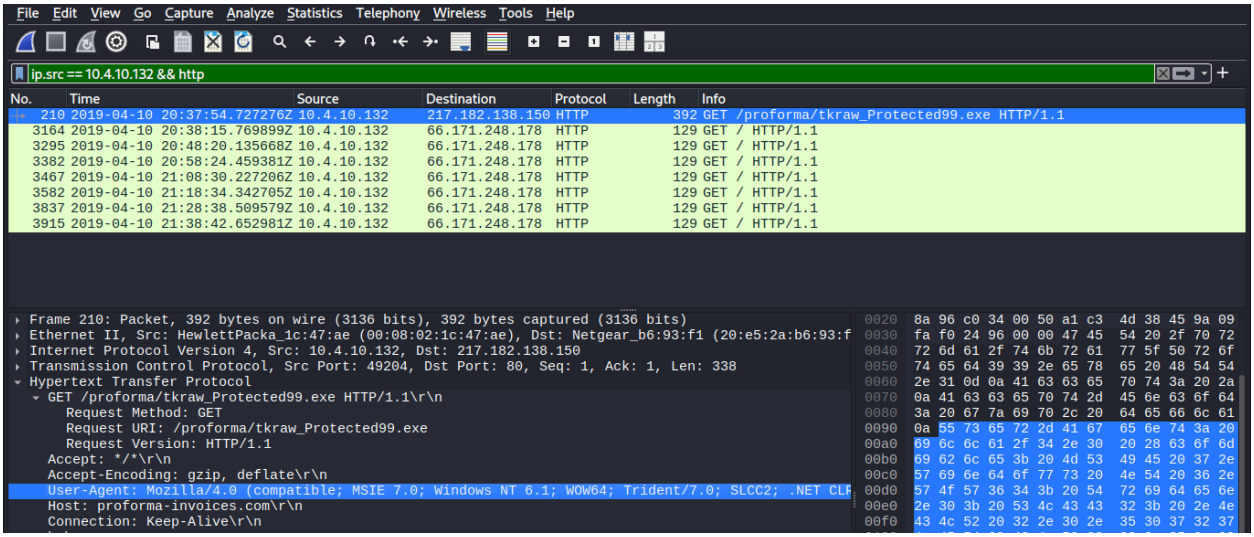
Analysis: Opened AbuseIPDB → Entered the IP address → Checked the country.



Q13. What operating system does the victim's computer run?

Answer: Windows NT 6.1

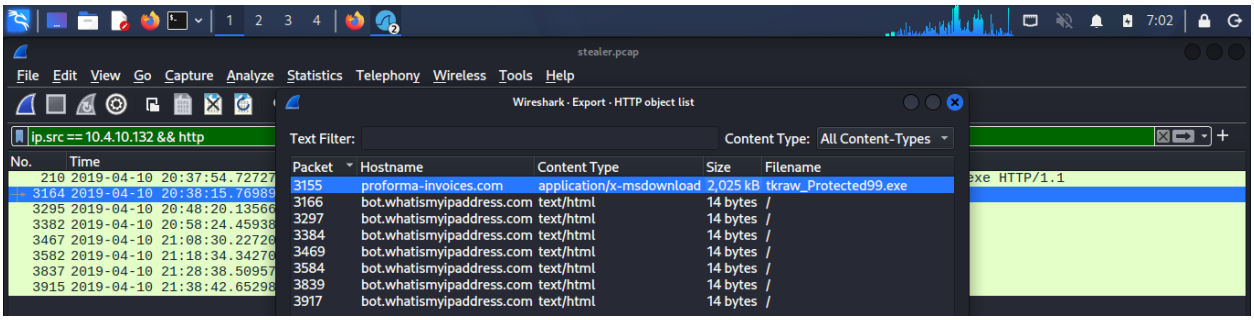
Analysis: Applied HTTP filter → Checked User-Agent → Identified the operating system.



Q14. What is the name of the malicious file downloaded by the accountant?

Answer: tkraw_Protected99.exe

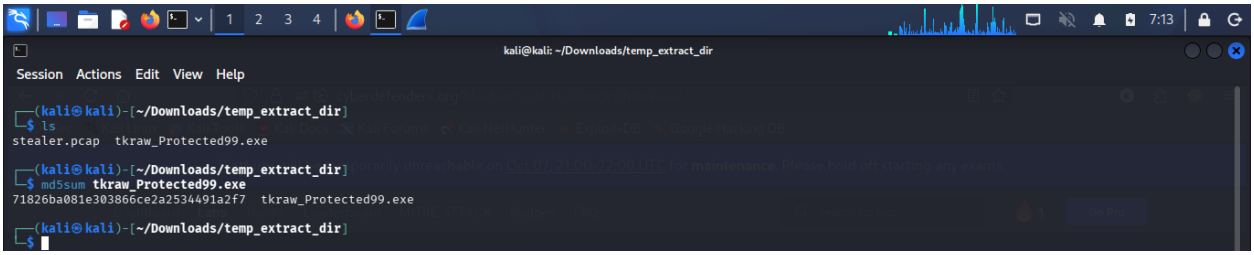
Analysis: File → Export Objects → HTTP → Found the downloaded executable.



Q15. What is the MD5 hash of the downloaded file?

Answer: 71826ba081e303866ce2a2534491a2f7

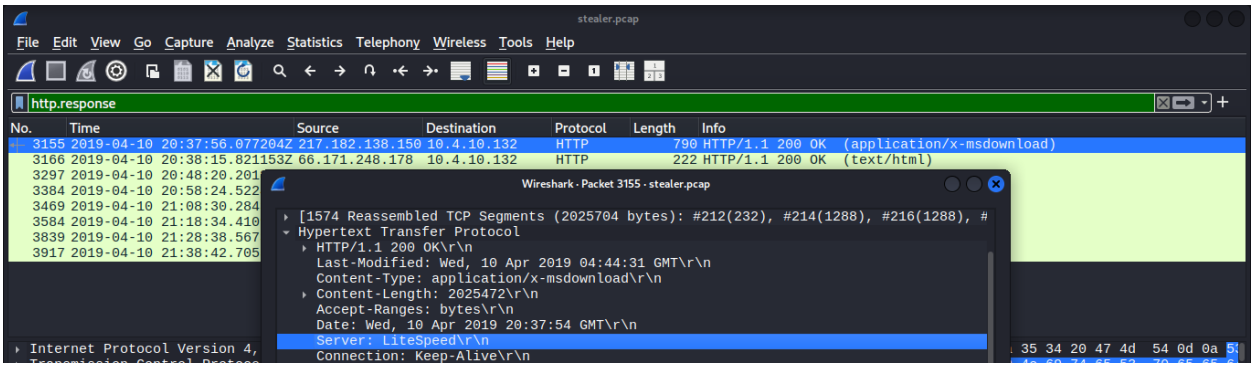
Analysis: Used the terminal `md5sum` command → Calculated the MD5 hash.



Q16. What software runs the web server that hosts the malware?

Answer: LiteSpeed

Analysis: Applied an `http.response` filter → Checked the response headers → Identified the server software.



Q17. What is the public IP of the victim's computer?

Answer: 173.66.146.112

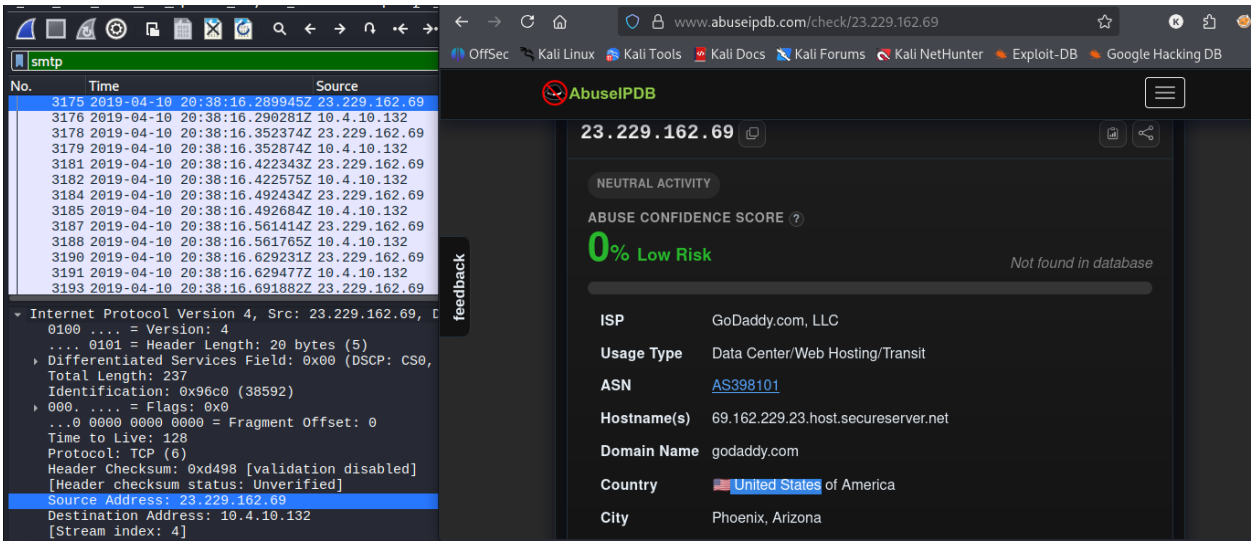
Analysis: Checked HTTP Response traffic → Followed HTTP Stream → Found the public IP address.



Q18. In which country is the email server to which the stolen information is sent?

Answer: United States

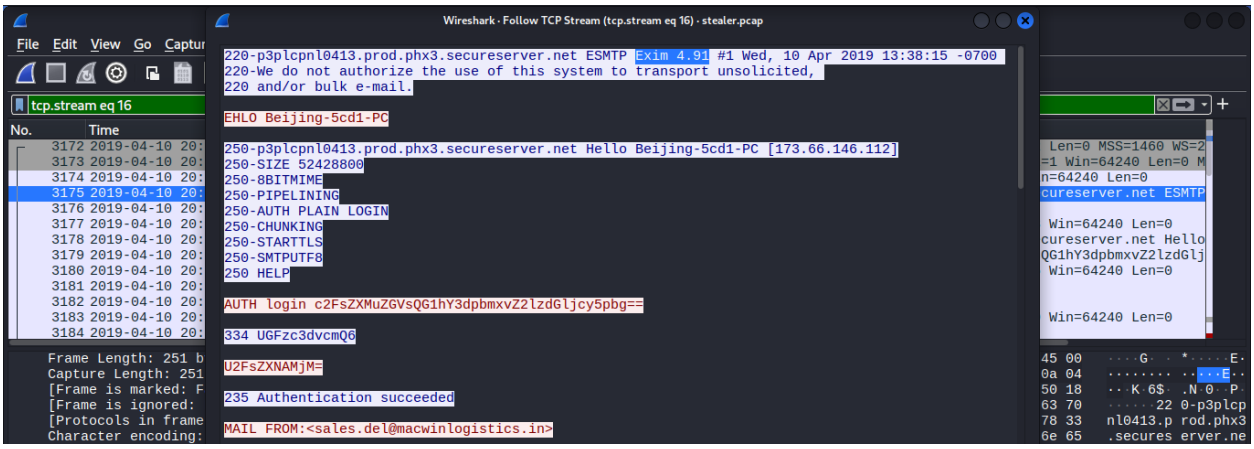
Analysis: Applied SMTP filter → Identified email server IP → Checked country using AbuseIPDB.



Q19. Analyzing the first extraction of information. What software runs the email server to which the stolen data is sent?

Answer: Exim 4.91

Analysis: Applied the SMTP filter → Followed the TCP stream → Checked the SMTP server response.



Q20. To which email account is the stolen information sent?

Answer: sales.del@macwinlogistics.in

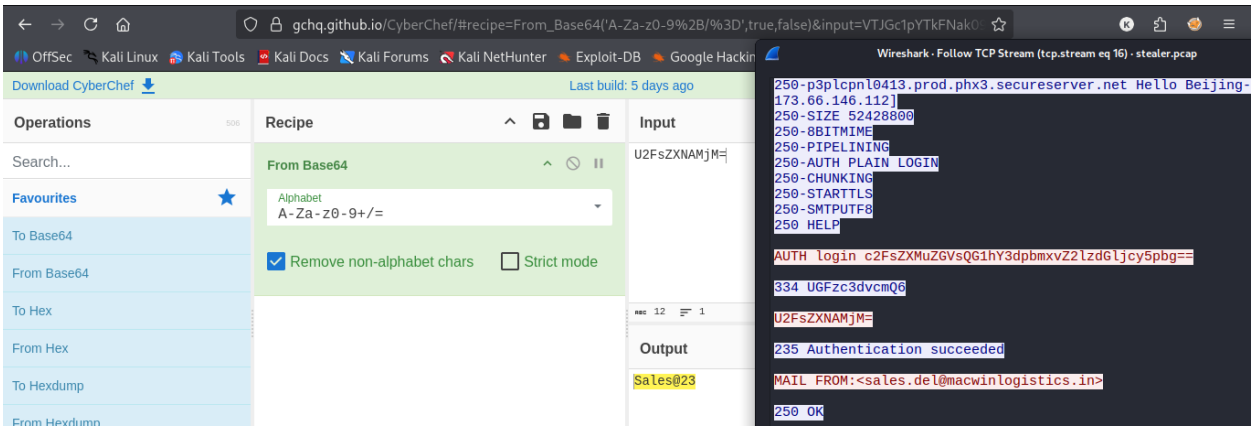
Analysis: Followed the SMTP TCP stream → Checked the recipient email address.



Q21. What is the password used by the malware to send the email?

Answer: Sales@23

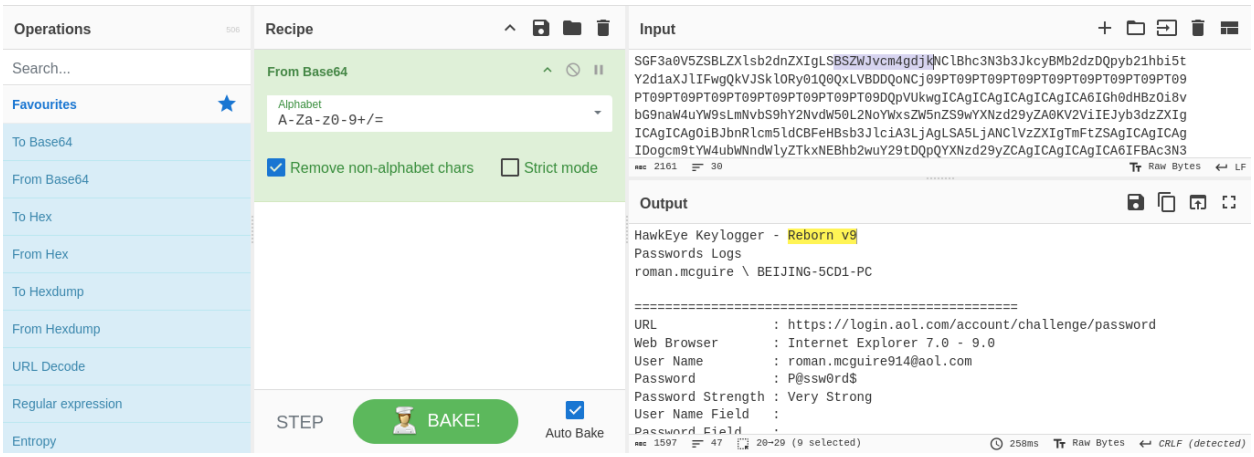
Analysis: Followed TCP Stream → Decoded Base64 authentication data using CyberChef.



Q22. Which malware variant exfiltrated the data?

Answer: Reborn v9

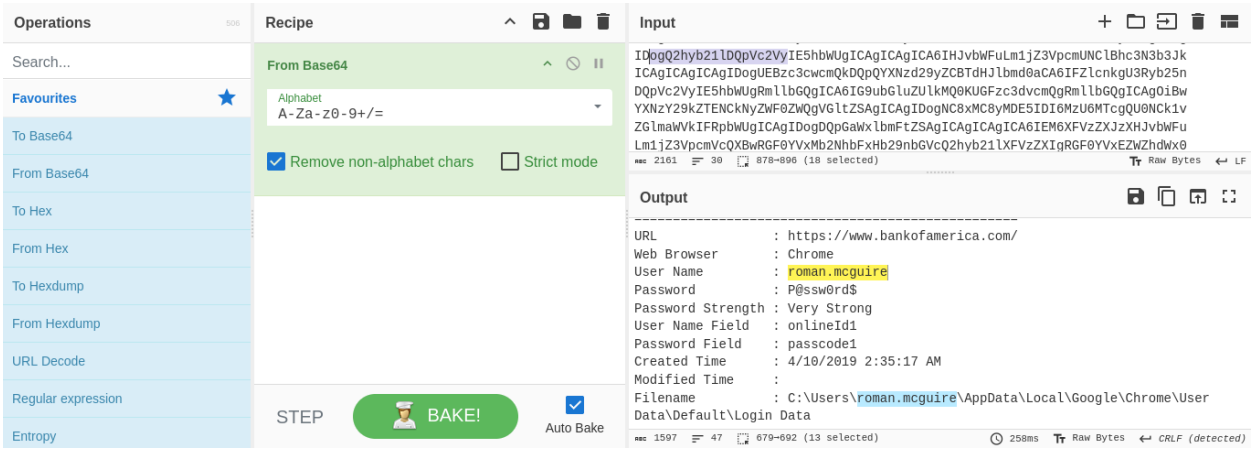
Analysis: Followed TCP Stream → Decoded Base64 email content using CyberChef → Identified the malware variant.



Q23. What are the bankofamerica access credentials? (username:password)

Answer: roman.mcguire:P@ssw0rd$

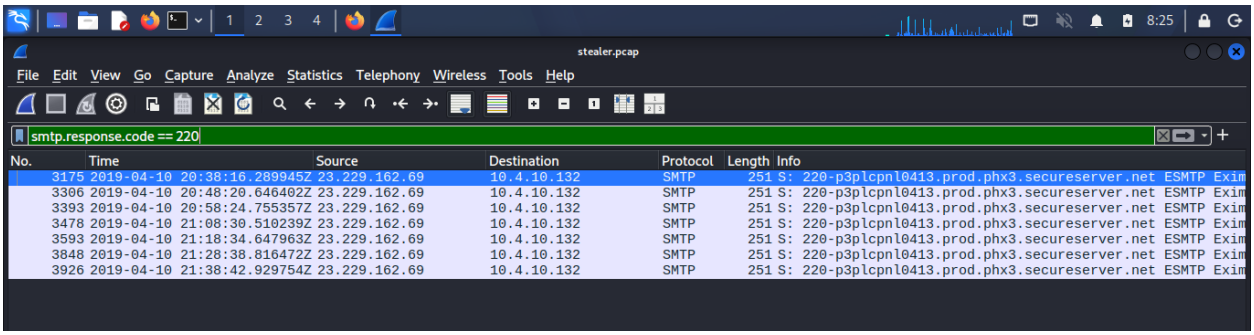
Analysis: Followed TCP Stream → Decoded email content → Examined the Bank of America data → Found the access credentials.



Q24. Every how many minutes does the collected data get exfiltrated?

Answer: 10

Analysis: Applied `smtp.response.code == 220` filter → Checked SMTP packet timestamps → Compared the intervals.



5. Conclusion

I completed the HawkEye Lab by analyzing the provided PCAP file in Wireshark and using CyberChef to decode encoded data.

I identified the network devices, DNS information, suspicious domain, downloaded executable, email server details, and evidence of credential theft. I also identified the Reborn v9 malware variant and determined that the collected information was sent approximately every 10 minutes.